

Neural bases of conditioned placebo analgesia

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ABSTRACT

Despite growing interest in the placebo effect, the neural correlates of conditioned analgesia are still incompletely understood. We investigated herein on brain activity during the conditioning and post-conditioning phases of a placebo experimental paradigm, using event-related fMRI in 31 healthy volunteers. Brief laser heat stimuli delivered to one foot (either right or left) were preceded by different visual cues, signalling either painful stimuli alone, or painful stimuli accompanied by a (sham) analgesic procedure. Cues signalling the analgesic procedure were followed by stimuli of lower intensity in the conditioning session, whereas in the test session both cues were followed by painful stimuli of the same intensity. During the first conditioning trials, progressive signal increases over time were found during anticipation of analgesia compared to anticipation of pain, in a medial prefrontal focus centered on medial area BA8, and in bilateral lateral prefrontal foci. These frontal foci were adjacent to, and partially overlapped, those active during anticipation of analgesia in the test session, whose signal changes were related to the magnitude of the placebo behavioral response, and those active during placebo analgesia. Specifically, a large focus in the right prefrontal cortex showed activity related to analgesia, irrespective of the expected side of stimulation. Analgesia was also related to decreased activity, detectable immediately following noxious stimulation, in parietal, insular and cingulate pain-related clusters. Our findings of dynamic changes in prefrontal areas during placebo conditioning, and of direct placebo effects on cortical nociceptive processing, add new insights into the neural bases of conditioned placebo analgesia.

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1. Introduction

The placebo analgesic effect is a neurobiological phenomenon that has been ascribed to different mechanisms, including expectation of relief from pain and conditioning [14].

Evidence from different clinical and experimental models, such as Parkinson's disease [4,12], hormone secretion [6], immune responses and emotion [33], suggests the crucial role of past experience for generating placebo effects. Clinical observations demonstrated the higher analgesic effectiveness of placebos when given after an active treatment [1,23]. The interactions between past experience and expectations of analgesia can be more thoroughly investigated in the experimental setting [11]. Behavioral studies in healthy volunteers indicate that placebo analgesia is stronger when preceded by a conditioning phase involving an effective analgesic procedure [10,29,39]; indeed, in an experimental protocol similar to ours [10], verbal instructions alone were ineffective in inducing placebo analgesia.

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Over the last few years, brain imaging studies have provided several insights into the neural mechanisms involved in placebo effects, namely on brain activity during the anticipatory phase and pain modulation, and on the involvement of the endogenous opioid system: however, discrepancies still exist as to the neural sites and timing of placebo-induced antinociceptive effects (see reviews in Refs. [5,19,38]). Moreover, knowledge on the neural mechanisms linking prior experience of effectiveness with the development of placebo analgesia is still limited [36].

A recent paper by Watson et al. [51] examined brain activity in a conditioned placebo analgesia protocol using a block-design fMRI paradigm, and showed a common brain network active during pain anticipation in the placebo conditioning and post-conditioning phases. Differently from Watson et al. [51], in the present study we adopted an event-related conditioning paradigm, which allowed us to investigate dynamic changes occurring over time during placebo conditioning and placebo analgesia. Our specific aims were to investigate whether cortical fMRI activity during anticipation of analgesia changed over time during the training session, whether overlapping foci were related to the learning phase and to the placebo analgesic effect in the Test run, and whether placebo was accompanied by a direct modulation of nociceptive activity, temporally unrelated to pain intensity rating. Moreover, we tested

two groups of subjects, stimulated on their right or left foot, in order to detect potential hemispheric asymmetries in placebo-related circuits. We focus herein on activity changes in the prefrontal cortex, which have been suggested to play a crucial role in conditioned placebo analgesia [5,13,49,51], as well as in a priori selected regions of the pain matrix.

2. Materials and methods

We investigated 36 healthy volunteers without any history of neurological or psychiatric illness, who were not under medications at the time of the study. Of the original 36 subjects, five were excluded from the study because of either excessive head motion ($n = 2$), or technical problems with the psychophysical recordings ($n = 1$), or with the MR acquisition ($n = 2$). The remaining 31 subjects (13 males, 18 females; mean age 23.5) entered the final analysis; among them, 26 were right-handed, 1 was left-handed and 4 were ambidextrous. Handedness was assessed using the Edinburgh Inventory [32].

All the experimental procedures were conducted in conformance with the policies and principles contained in the Declaration of Helsinki, and all subjects gave their written informed consent to take part in the study. Volunteers were told that the aim of the study was to assess both the efficacy and the brain correlates of a new analgesic procedure; to this end, they would receive noxious cutaneous heat stimuli on the top of one foot and, in some trials, the painful stimulus would be accompanied by a sub-threshold electric stimulation at the ankle, which could induce analgesia. Subjects received a small remuneration for taking part in the experiment. All the subjects were debriefed at the end of the study: none of them showed awareness of the placebo procedure, and the subjects who reported as feeling more pain in the last session ascribed this occurrence either to sensitization or to loss of effectiveness of the supposedly administered analgesic procedure.

2.1. Laser stimuli

Brief radiant heat pulses, generated by an infrared neodymium:yttrium-aluminium-perovskite (Nd:YAP) laser stimulator (Electronical Engineering, Florence, Italy) with a wavelength of 1.34 μm , were used as noxious or non-noxious stimuli. At this short wavelength, the laser pulses directly activate A δ - and C-fiber terminals located in the superficial layers, without inducing damage to the skin [16]. The laser beam was focused by a converging lens to reach a diameter of approximately 10 mm; its energy range was 0.5–10 J; duration 1–10 ms. A visible He–Ne pilot laser was used to point to the area to be stimulated. To avoid nociceptor fatigue or sensitisation, the laser beam was moved randomly after each stimulus over a 3×5 cm skin area.

2.2. Experimental session

The experimental session started with the assessment of the individual responses to laser stimulation. An ascending series of stimuli were delivered in steps of 0.5 J, starting from very low intensities (0.5 J, below warmth threshold) until a mild-to-moderate pain intensity was achieved for each subject. During the subsequent experimental phase, the intensity was set either at a level inducing a non-painful (N) warm sensation, just below pain threshold, or at a level inducing pain (P) of moderate intensity. In each subject, only one foot was stimulated. The side of stimulation was randomized across subjects: the right foot was stimulated in 15, the left foot in 16 volunteers.

Two conditioning runs and one Test run, each including 12 trials (Fig. 1A), were carried out for each subject while fMRI images were acquired. The interval between subsequent runs was 2–3 min.

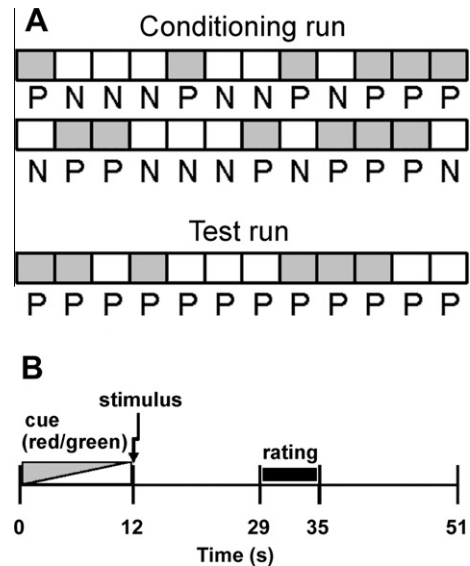


Fig. 1. Experimental design. (A) Volunteers underwent three runs of 12 trials each. In each run, Red ($n = 6$) and Green ($n = 6$) trials were pseudo-randomly alternated, with a maximum of three consecutive trials of the same type. (B) Time course of events in each trial. N, non-painful stimulation; P, painful stimulation. Grey and white squares represent Red and Green trials, respectively.

Each trial lasted 51 s (Fig. 1B). At time 0, volunteers received a visual warning cue, i.e., the black screen they were looking at turned either Red ("Red" trials) or Green ("Green" trials). Later (12 s) the screen turned black again, and immediately afterwards the subjects received a laser stimulation on the top of one foot. The volunteers had been informed that the Red cue would be followed by a brief painful laser stimulus, whereas the Green cue would be followed by an identical painful stimulus associated to a sub-threshold electric shock, which could induce analgesia (the placebo manipulation). To this purpose, two electrodes were pasted above the ankle; however, the electrodes were not connected to any pulse generator, and no electric shock was ever delivered. In the two conditioning runs, the laser stimulus intensity after the Green cue was milder (non-painful stimulus, N), with respect to the stimulus intensity following the Red cue (painful stimulus, P). On the contrary, in the Test run the stimulus following the Green cue had the same power as the stimulus following the Red cue (painful stimulus, P). We did not randomize the cues across subjects, because colors are known to impact on placebo responses (see Section 4).

Seventeen seconds after the stimulus, volunteers had to rate the perceived pain intensity, by rotating a knob which moved a cursor on a computerized visual analogue scale (VAS), anchored at 0 = no pain, and 100 = worst imaginable pain. Volunteers were accustomed to the rating procedures before the actual fMRI experiments. The individual differences in pain ratings between the Red and the Green trials in the Test run were taken as a measure of the placebo effect.

Visual stimuli were rear-projected on a screen placed in front of the gantry, and were made visible to the volunteers by means of a pair of angled mirrors located above the head-coil. Behavioral responses were collected by means of a custom-made, MR-compatible device, and software developed in Visual Basic 6 (http://digilander.libero.it/marco_serafini/stimoli_video/).

2.3. Statistical analysis of psychophysical responses

Data were analyzed using repeated-measures ANOVA, after testing normal distribution with the Kolmogorov–Smirnov test. Sphericity condition was assessed and when it was not verified,

the Greenhouse–Geisser correction was applied. ANOVA included the following within-subjects factors: cue (Red vs. Green), runs, and trial order. The *F*-tests were followed by the Bonferroni *post hoc* tests for multiple comparisons.

All the analyses were carried out using the SPSS software package (SPSS Inc, Chicago, Illinois, USA). The level of significance was set at $p < 0.05$.

2.4. fMRI data acquisition and analysis

Functional images were acquired with a Philips Intera MR system at 3T and a gradient-echo echo-planar sequence (TR = 3000 ms; TE = 35 ms; FOV = 240 mm; in-plane acquisition matrix 80×80 , reconstructed at 128×128 ; 30 axial slices; voxel size: $1.9 \times 1.9 \times 3.5$ mm, with a 0.5 mm gap between contiguous slices). High resolution T1-weighted anatomical images were acquired for each subject to allow anatomical localization (TR = 9.9 ms; TE = 4.6 ms; 170 sagittal slices; voxel size = $1 \times 1 \times 1$ mm).

The Matlab 7.1 (The MathWorks, Inc.) and SPM5 (Wellcome Department of Imaging Neuroscience, London, UK) softwares were used for data analysis. For each subject, all functional volumes were realigned to the first volume acquired, slice time corrected, normalized to the MNI (Montreal Neurologic Institute) template implemented in SPM5, and smoothed with a $4 \times 4 \times 8$ mm FWHM Gaussian kernel.

The analysis of fMRI data was performed under the framework of the general linear model. Separate regressors, related to the anticipation or to the stimulus perception phases, were defined for each run and for each group of trials (Red or Green). The stimuli were modelled as brief (1 TR) events, whereas anticipation was modelled as a block lasting from cue onset to actual stimulation (4 TRs). The rating period was also included in the analysis matrix as a regressor.

In each subject, condition specific effects were compared using linear contrasts. To test for potential changes over time during training, the anticipation phases were analyzed in the two conditioning runs using a linear regression with trial order; because unpublished data from our group suggest that placebo learning already occurs after few trials (Colloca et al., personal observations), separate analyses were performed for Run 1 and Run 2.

Trials in the Test run were analyzed using a linear regression with trial order, with the subject's psychophysical rating, or with the combination of the two. The individual contrast images were then submitted to a second-level random-effect analysis with separate one-sample *T*-test. In order to map regions whose activity was related to analgesia, an individual estimate of the placebo analgesic response (the *t* value obtained for each participant from the comparison of pain intensity values reported in the Red vs. Green trials during the Test run) was also entered as a regressor in the second-level analysis. A double statistical threshold (voxel intensity and spatial extent) was adopted to achieve a combined significance, corrected for multiple comparisons, of $\alpha < 0.05$, as assessed by AlphaSim (Alpha Simulations) with 1000 Monte Carlo simulations (<http://afni.nimh.nih.gov/afni/doc/manual/AlphaSim>).

To test the hypothesis of a placebo-induced reduction of brain responses to the painful stimulus [13,37,49,51], we compared signal changes following noxious thermal stimulation in the Red vs. Green trials of the Test run, in parietal, insular and cingulate cortical regions which have been consistently shown to be active during thermal pain perception [2]. These same regions also showed a robust pain-related fMRI signal increase in our experiments during the Red trials (data not shown). Small volume corrections (SVC) analyses were performed on spheres of interest (8 mm radius), whose centers were chosen on the basis of the coordinates reported in previous fMRI studies which investigated the brain responses to thermal noxious stimulation of the foot in healthy

human volunteers [3,7,8,30,31,46]. Because noxious-evoked responses in the parietal operculum, cingulate and insular cortex are often bilateral [2,35], we performed the analysis on the whole group of 31 volunteers, even if they received noxious stimulation on to one foot only. To analyze placebo-induced differences in the time profiles of fMRI signals following noxious stimuli, individual adjusted peristimulus time-courses (PST) were averaged from trials of the Test run for each condition (Red or Green), and were compared using repeated-measures ANOVA (with cue and time as the within-subjects factors) and simple contrasts.

Coordinates in the Talairach space were obtained by applying the M. Brett algorithm (mni2tal: <http://www.mrc-cbu.cam.ac.uk/Imaging/mnispace.html>). Brodmann areas were identified with the help of the MSU utility (http://www.ihb.spb.ru/~pet_lab/MSU/MSUMain.html).

Because a direct between-groups comparison of fMRI data from volunteers stimulated on their right or left foot revealed no significant foci in all considered conditions, with the only exception of a lateral orbitofrontal cluster more active in the Left than in the Right group during the stimulus perception phase in Green trials of the Test run (see [Supplementary Fig. 1](#)), we describe the results from the whole group of 31 volunteers.

3. Results

3.1. Psychophysical data

Repeated-measures ANOVA on the VAS scores showed a main effect for cue ($F_{(1,28)} = 92.01$, $p < 0.0001$) and run ($F_{(2,56)} = 5.90$, $p < 0.02$). A significant interaction between the two factors was found ($F_{(2,56)} = 30.08$, $p < 0.0001$), indicating variability across the three experimental runs, as expected.

In the two training runs (1 and 2), a difference was found between VAS ratings of Red- and Green-associated stimuli, as expected [main effect for cue ($F_{(1,30)} = 93.53$, $p < 0.0001$); run ($F_{(1,30)} = 0.11$, $p = 0.73$)].

A separate analysis demonstrated that, also in the Test run, subjects rated noxious stimuli as consistently less painful in Green trials, namely, when they expected an analgesic effect (cue: $F_{(1,28)} = 26.62$, $p < 0.0001$). Considering the whole group, the placebo effect accounted for an approximate 33% average reduction in VAS pain ratings ([Fig. 2](#), top). Trial order did not meet significance: $F_{(5,140)} = 2.25$, $p = 0.07$ ([Fig. 2](#), bottom). At the individual level, the reduction was significant in 12 subjects only ("placebo responders"). Because we did not find a bimodal, but rather a continuous distribution of average placebo-induced VAS changes, we analyzed the fMRI data using an estimate of the placebo effectiveness as regressor (see [Section 2](#)) on all 31 subjects.

3.2. fMRI data

3.2.1. Training runs

In order to detect regions showing build-up of activity during the training runs, and thus potentially involved in placebo conditioning, we first compared signal changes during the anticipation phase when volunteers expected an analgesic effect (Green trials), with signal changes related to the expectation of pain (Red trials); each condition was analyzed parametrically according to the succession of the 6 trials, separately for Run 1 and Run 2 (see [Section 2](#)). The Green anticipation vs. Red anticipation comparison for trials of Run 1 revealed a medial prefrontal (MPFC) focus centered on medial BA8, and bilateral lateral prefrontal foci ([Table 1](#) and [Fig. 3](#)). In addition to frontal foci, the same contrast revealed significant active foci in several parietal, occipital and cerebellar regions ([Supplementary Fig. 2](#)). In Run 2, a similar comparison of Green anticipation vs. Red anticipation did not reveal significant

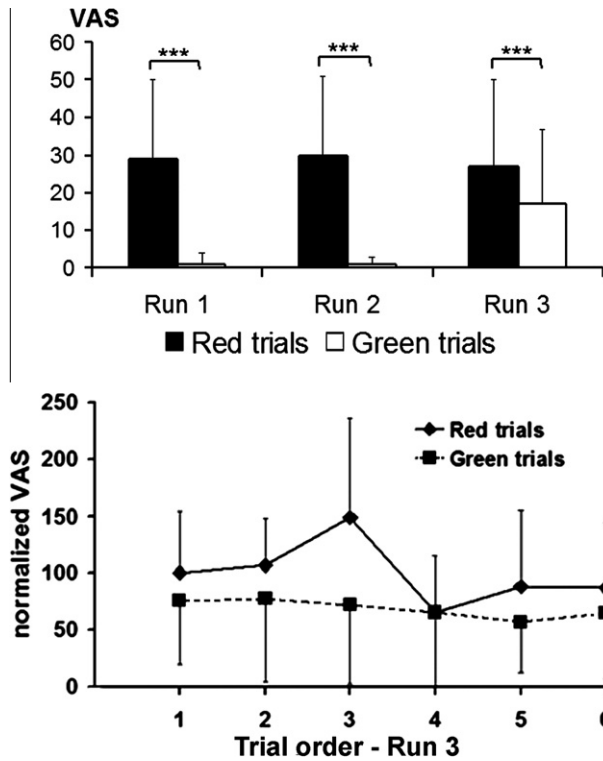


Fig. 2. Psychophysical data. Top: Average (+SD) VAS scores in Red and Green trials of the three runs. ***Significant differences at $p < 0.001$ (paired- t test). Bottom: psychophysical ratings in the Test run; no significant effect was found depending on trial order. VAS data are normalized to the average of the 6 Red trials (=100).

signal changes. The early build-up of brain activity during the training phase is confirmed by another analysis, in which we directly compared Green anticipation vs. Red anticipation in Run 1 vs. Run 2: the results showed indeed several foci in medial, middle and inferior frontal gyri, whose changes over time during the anticipation phase were higher during the first than during the second conditioning run (Supplementary Table 1).

3.2.2. Test run: Anticipation phase

In order to assess which regions showed activity related to the degree of the placebo analgesic response during anticipation of analgesia, we compared Green anticipation vs. Red anticipation in the Test run, using an estimate of the placebo behavioral response (see Section 2) as linear regressor. A large active focus was found in the right prefrontal cortex (BA9/46) (Table 2A; Fig. 4, top row). None of the extra-frontal foci identified during the anticipation phase in the first training run (see above) showed significant placebo-related changes during the anticipation phase in the Test run.

To assess potential changes over time in prefrontal regions during the Test run, Green anticipation and Red anticipation were

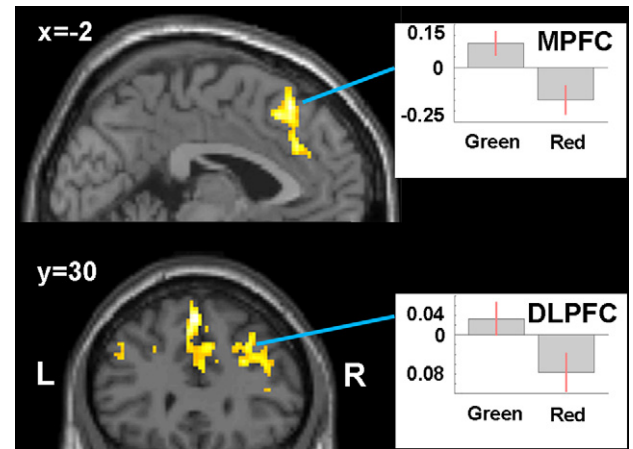


Fig. 3. Brain regions related to anticipation of analgesia during the first conditioning run: foci showing higher activity during the anticipation phase in Green trials (when subjects expected analgesia) than in Red trials (when subjects expected pain), whose signal increases were linearly related to trial order. Histograms represent the plots of parameter estimates (mean $\pm$ SEM) for Green and Red trials: both foci in the medial (MPFC) and dorso-lateral (DLPFC) prefrontal cortex are characterized by increase vs. decrease of parameter estimates, in Green vs. Red trials, respectively. L: left side; R: right side. x and y values represent the medio-lateral and antero-posterior coordinates, respectively, of planes in the MNI space.

compared including trial order as a parameter in the analysis. We found lateral prefrontal clusters, whose location (more rostral and ventral) was clearly different from the ones described above, whereas a MPFC focus partially overlapped the focus found during anticipation in the first conditioning run (Table 2B; Fig. 4, bottom row). An additional parametric analysis, using the combination of trial order and subject psychophysical rating, did not reveal any significant signal change.

3.2.3. Test run: Stimulus perception phase

Medial and lateral prefrontal foci showed higher activity in Green trials than in Red trials (Table 2C); they were adjacent to, and partly overlapped, the frontal foci found during anticipation of analgesia in the first conditioning run and in the Test run (Figs. 5 and 6).

To test whether the placebo analgesic response was accompanied by a reduction of brain responses to the painful stimulus, we analyzed a priori selected regions of the pain matrix. The results showed that during the stimulus perception phase there were lower changes in the Green trials than in the Red trials, parametrically related to the magnitude of placebo analgesia, in the left parietal operculum, posterior insula, the right middle insula, the pMCC and aMCC. The analysis of peristimulus time courses in all 31 subjects revealed that placebo-induced decreases in the activity of the pain matrix occurred during the stimulus perception phase (Fig. 7 and Supplementary Table 2). Interestingly, only in the aMCC we found a cue $\times$ group interaction when comparing data from placebo responders and non-responders (see Supplementary Table 2 and Supplementary Fig. 3).

Table 1

Conditioning Run 1: anticipation of analgesia vs. anticipation of pain. Regions of increased signal, parametrically analyzed according to the sequence of the 6 trials of Run 1.

	Cluster level		Voxel level			MNI coord. (mm)			Talairach coord. (mm)		
	k	p (corr)	t	Z	p	x	y	z	x	y	z
L/R medial frontal gyrus, anterior middle cingulate cortex	485	<0.001	5.57	4.56	<0.001	2	34	28	2	34	24
L middle and superior frontal gyrus	231	<0.001	4.86	4.12	<0.001	-24	32	38	-24	33	33
L middle and superior frontal gyrus	216	<0.001	4.67	4.00	<0.001	-30	48	14	-30	47	11
R middle and inferior frontal gyrus	660	<0.001	4.65	3.99	<0.001	26	24	32	26	25	28

R = right, and L = left hemisphere. k = Voxel number.

Only foci located in the frontal lobe which reach the combined voxelwise/clusterwise significance level of $\alpha < 0.05$ (corr.) are considered.

Table 2

Test run: Green trials vs. Red trials.

	Cluster level		Voxel level			MNI coord. (mm)			Talairach coord. (mm)		
	k	p (corr)	t	Z	p	x	y	z	x	y	z
<i>A. Region of increased signal in anticipation of analgesia vs. anticipation of pain, using the individual degree of placebo response as a regressor</i>											
R middle and inferior frontal gyrus	201	<0.001	6.03	4.82	<0.001	44	30	26	44	30	22
<i>B. Regions of increased signal in anticipation of analgesia vs. anticipation of pain, parametrically analyzed according to the succession of the 6 trials in time</i>											
R middle and superior frontal gyrus	383	<0.0001	5.40	4.46	<0.001	34	58	8	34	57	5
R inferior frontal gyrus	143	0.004	4.96	4.18	<0.001	32	24	−14	32	23	−13
R–L medial frontal gyrus, cingulate gyrus	132	0.007	3.92	3.48	<0.001	6	26	38	6	27	34
<i>C. Regions of increased signal during pain perception, in Green vs. Red trials of the test run</i>											
R middle and inferior frontal gyrus	663	<0.001	6.13	4.87	<0.001	52	20	34	51	21	30
R medial and superior frontal gyrus	751	<0.001	5.91	4.75	<0.001	6	54	16	6	53	12
R superior and middle frontal gyrus	149	0.002	4.70	4.02	<0.001	30	54	−2	30	52	−4

R = right, and L = left hemisphere. k = Voxel number.

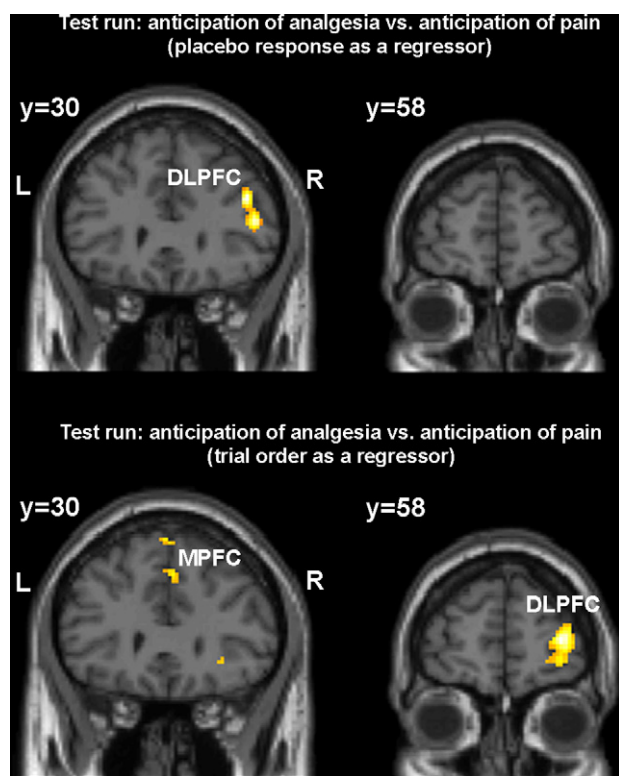
Only foci located in the frontal lobe which reach the combined voxelwise/clusterwise significance level of $\alpha < 0.05$ (corr.) are considered.

Fig. 4. Brain regions showing placebo-related fMRI signal increases during the anticipation phase in the Test run. Top: Frontal focus showing higher activity during the anticipation phase in Green trials (when subjects expected analgesia) than in Red trials (when subjects expected pain), whose signal increases were positively related to the individual placebo response. Bottom: Frontal foci showing higher activity during the anticipation phase in Green than in Red trials, whose signal increases were positively related to the trial order. L: left side; R: right side. y values: antero-posterior coordinates of planes in the MNI space.

4. Discussion

To our knowledge, this is the first study describing dynamic neural changes during a learning process for placebo analgesia. We demonstrate that, during training, anticipation of analgesia is accompanied by signal increases over time in medial (MPFC) and lateral (DLPFC) prefrontal foci. DLPFC foci partly overlap with a large focus in the right lateral prefrontal cortex, which is active when subjects expect an analgesic treatment in the test session, and whose activity is related to the degree of placebo analgesia. These frontal foci are likely involved in the development of placebo analgesia.

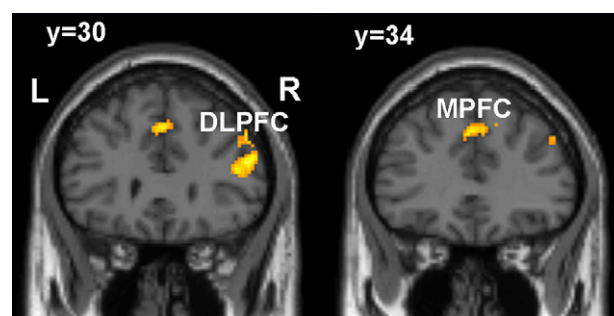


Fig. 5. Brain regions showing placebo-related fMRI signal increases during the stimulus perception phase in the Test run: medial and lateral prefrontal foci showing higher activity in Green trials than in Red trials during the stimulus perception phase. L: left side; R: right side. y values: antero-posterior coordinates of planes in the MNI space.

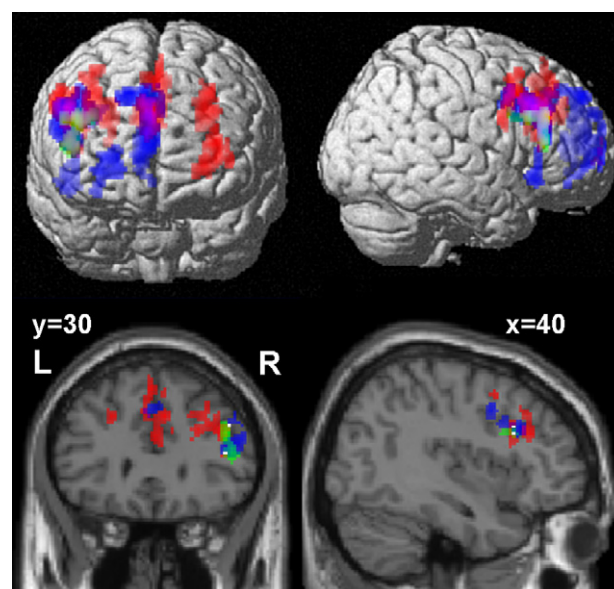


Fig. 6. Regions of fMRI signal increases induced by placebo during pain perception partly overlapped with regions of signal increases during anticipation of analgesia in the first conditioning run and in the Test run. Top: three-dimensional rendering, frontal view (on the left) and right view (on the right). Bottom: coronal section (on the left) and sagittal section (on the right). Red: anticipation of analgesia in the first conditioning run; Green: anticipation of analgesia in the Test run; Blue: placebo-induced signal increases during pain perception; White: overlap of all three maps. L: left side; R: right side. x and y values: medio-lateral and antero-posterior coordinates (respectively) of planes in the MNI space.

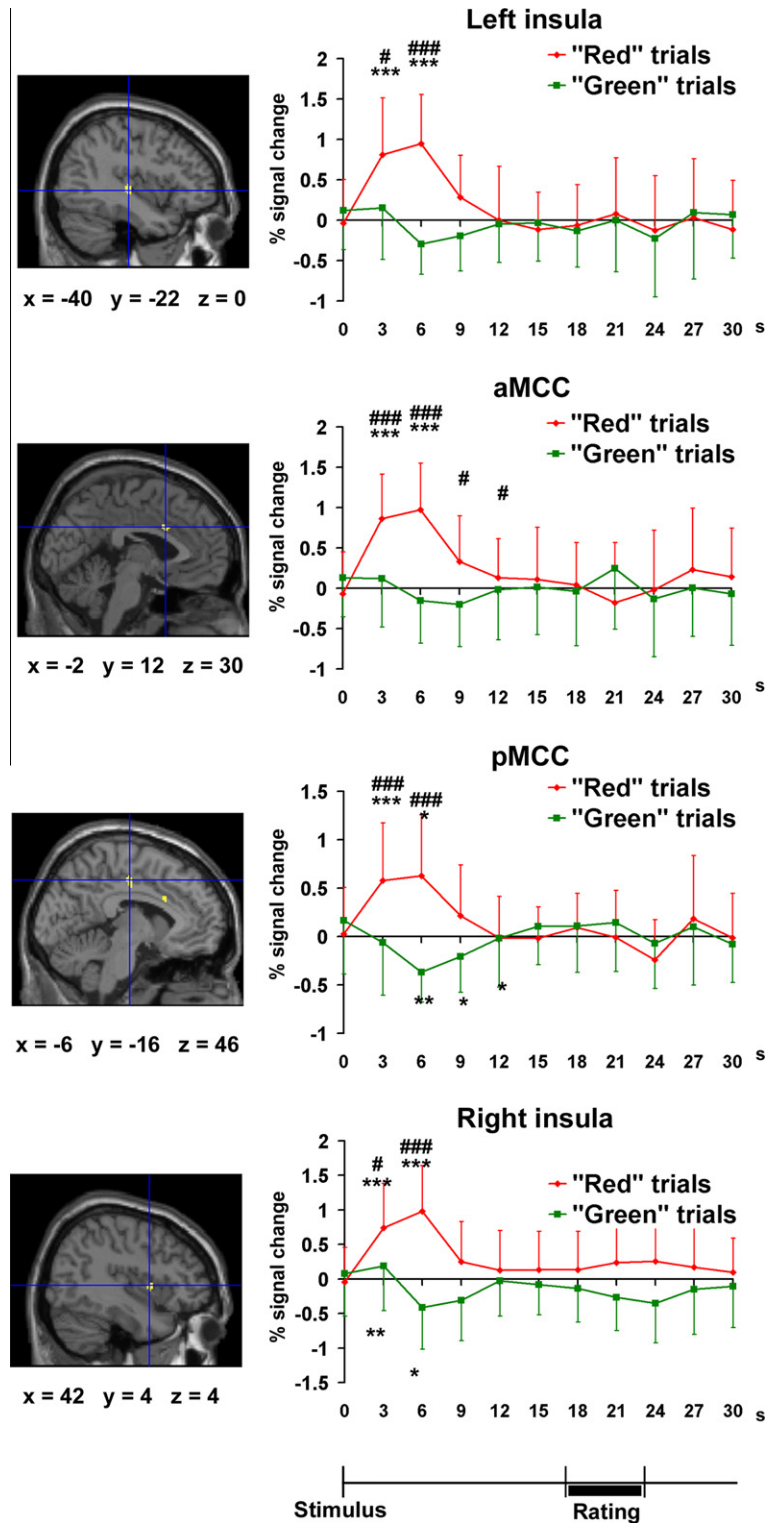


Fig. 7. Time courses of placebo-induced modulation of pain-related signals. Peristimulus time courses of percentage fMRI signal changes (y axis) in peak voxels of pain-related clusters during the Red and Green trials of the Test run. Data are mean ($\pm$ SD) from all 31 subjects included in the random-effect analysis. *, ** and ***: significantly different from pre-stimulus values, ANOVA + simple contrasts, $p < 0.05$, $p < 0.005$, and $p < 0.001$, respectively. # and ###: significantly different from Green trials, paired t -tests, $p < 0.05$ and $p < 0.001$, respectively.

Moreover, we provide evidence for placebo-induced modulation of nociceptive activity during the stimulus perception phase. These results suggest, therefore, that differential prefrontal responses to the cues may reflect processes that affect pain perception and not simply pain reports, confirming and extending previous findings [13,22,34,37,49].

4.1. Development of the placebo effect: anticipation phase in the conditioning and test sessions

Understanding anticipatory activity in the conditioning session is an important step, both theoretically and clinically, in explaining

how externally reinforced stimuli are modulated to induce learned placebo analgesic responses [17].

We did not randomize cue colors across subjects, because colors are part of our designs. In fact, it has long been known that colors of pills impact on placebo responses. As individuals show a trend to associate Red to hazard and avoidance motivation [27], this might have further contributed to different expectations following the Red or the Green cues.

We found active foci in several posterior brain regions during the anticipation phase in the first training run (Supplementary Fig. 2); however, none of them showed significant placebo-related changes during the anticipation phase in the Test run, thus they might be generally related to conditioning mechanisms. For instance, activity in occipito-parietal areas might be due to attentional mechanisms related to the salience of visual stimuli related to the conditioning process; the cerebellum has been implicated in aversive conditioning [41], but also in placebo conditioning [18,21]. As one of the goals of our study was to describe common areas of activation during the conditioning and post-conditioning sessions, we focussed our attention on frontal areas, which showed activity changes in both these conditions and are thus likely to be more directly related to anticipation of analgesia. Indeed, the pattern of prefrontal activation described herein is quite different from the brain pattern of activity typically associated with other conditioning models in humans, e.g. fear conditioning and extinction, which involve mainly the amygdala, anterior cingulate and insular cortex [9,44].

Our results agree with those of Watson et al. [51], in that we show an overlap of anticipation of analgesia-related activity in the frontal cortex, during the conditioning and post-conditioning sessions. Furthermore, our event-related design allowed us to extend their previous findings, detecting dynamic prefrontal activity changes in the conditioning session, namely a build-up over time of fMRI signal changes related to anticipation of analgesia, which was evident during the first conditioning trials. These findings are in accordance with behavioral data from our group (Colloca et al., unpublished) showing that, even after a small number of conditioning trials (although not after a single trial), long-lasting placebo analgesia may occur. In different paradigms involving a larger number of trials, the dynamics of conditioning of pain relief in humans was found to be non-linear (e.g. [45]).

4.2. Role of the dorsolateral prefrontal cortex in placebo-induced modulation

An additional new finding of the present study is that fMRI signal increases occur in the same prefrontal areas during anticipation of analgesia and during the stimulus perception phase in the test session (namely, when placebo analgesic effects occur). Moreover, differently from Watson et al. [51], we observed a clear right hemisphere predominance of placebo-related activity in the lateral prefrontal cortex, irrespective of the expected side of stimulation. The discrepancies in the location of active foci between the results of the two studies could be due to the different experimental protocol and data analysis. Indeed, as previously pointed out [36], given the complexity of factors potentially involved in placebo learning and analgesia, it is likely that different paradigms can involve activations of different cortical circuits. Another recent paper [22] did not reveal side differences in the placebo modulation obtained by TMS stimulation of DLPFC. Nevertheless, we investigated a rather large group of volunteers, stimulated either on the right or on the left foot; therefore, our results point to a crucial role of right DLPFC in conditioned placebo analgesia.

DLPFC is known to be involved in perceptual and cognitive control [28]; specifically, both medial and lateral (especially on the

right) prefrontal cortices are involved in the cognitive regulation of the pain system [24,34,42,49,52,53].

The right lateral prefrontal cortex may be related to the integration of expectation-related information with afferent information [21] and to reflective conscious responses to negative affect [24,53]. Activity in a right DLPFC focus, whose peak coordinates are almost identical to those described herein, is related to the perceived intensity of suggestion-induced pain under hypnosis [40]. It has been proposed that the dorsolateral prefrontal cortex (DLPFC) exerts active control on pain perception by modulating cortico-cortical pathways [25]. Interestingly, transcranial magnetic stimulation of the right, but not of the left, DLPFC increases pain tolerance in the cold pressor test during stimulation of either hand [15].

Therefore, we hypothesize that a building up of activity in the right PFC is critically involved in setting the volunteers' conditioning-induced expectancies about analgesia [29,39], thus leading to decreased pain perception. Interestingly, anticipation of an analgesic benefit is characterized by an increase in opioid activity in the right PFC, including DLPFC, and in pregenual ACC [50], a key area for pain affect and vegetative control [47]. A recent study showed that the activation of the DLPFC and pregenual ACC predicted the placebo-induced reduction of BOLD responses in the pain matrix, and was reduced by naloxone [13], further demonstrating the involvement of opioidergic mechanisms in conditioned placebo analgesia.

Given the adopted experimental paradigm, we could not look for placebo-related behavioral changes during the training runs, but did so in the Test run: during the 6 test trials, we did not find evidence for a decrease in the placebo analgesic effect, or in analgesia-related changes in the prefrontal cortex, despite the mismatch between the expected and the actual afferent input. However, we found signal changes over time, during the Test run, in medial and inferior lateral regions of the prefrontal cortex. These regions are clearly different from the DLPFC foci that we have described so far, and might be involved in the mismatch detection, but do not appear to cause a modulation in actual pain perception. Therefore, additional studies, covering more trials, are needed to assess the time course of, and the mechanisms underlying, the waning of the placebo analgesic effect.

4.3. Mechanisms and sites of placebo-induced modulation of the pain matrix

A crucial issue in the interpretation of placebo analgesia is whether it reflects actual reduction of nociceptive processing, or simply rating biases [38]. The landmark fMRI study by Wager et al. [49] showed reduction in neural activity of pain-sensitive regions (such as the insular and cingulate cortex) only after stimulus offset and during the rating period; therefore, the observed neural changes could have been potentially related to modulation of the cognitive evaluation of pain intensity, which involves activity in these regions (see Refs. [20,26]).

We show herein that placebo-induced decreases in pain-related functional responses were detectable during the stimulus perception phase, not just when the subjects rated the pain. Thus, our results in healthy volunteers confirm and extend those of a clinical fMRI study in patients suffering from irritable bowel syndrome [37] showing that placebo-induced reductions in brain activation occurred while they experienced pain. Differences with the results by Wager et al. [49] might be ascribed to the different experimental paradigm (brief laser stimuli vs. prolonged contact heat), in that ours may facilitate the detection of the time courses of pain- and analgesia-related activity.

Our findings, together with those of Price et al., lend strong support to the hypothesis that placebo analgesia is based upon

changes in actual processing of nociceptive information and, therefore, upon changes in pain perception [38], although additional components cannot be ruled out [48,49].

As in previous studies [18,49], we found a relationship between the placebo-induced fMRI signal changes in specific cortical regions and the subjective analgesic reports. Individual differences in the magnitude of opioids [50] and dopamine [43] release following placebo manipulations have also been described in the literature. It may be hypothesized that, by combining different brain mapping approaches, we may achieve a better understanding of the mechanisms accounting for inter-individual variability in the placebo analgesic responses, which might prove of great importance for clinical applications.

In summary, our data show that a common prefrontal network takes part in the different phases of learned analgesia, and might be a determinant factor for setting the power of placebos. Specifically, signal changes during anticipation in lateral prefrontal regions appear to be an important initiator of, and to play a crucial role in, placebo-induced modulation of pain networks involved in pain perception.

Conflict of interest statement

The authors declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at [doi:10.1016/j.pain.2010.09.021](https://doi.org/10.1016/j.pain.2010.09.021).

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